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ASSETS TRACKED	OBSERVATIONS	TIME HORIZON	SHOCK REGIMES
8	6,410	2000-2025	3

CRASHVIZ

The Geometry of Regimes

Visualizing market connectivity and systemic risk across 25 years of daily closes.



Welcome page -- the live regime classifier reads ELEVATED STRESS on the most recent correlation matrix.

From Idea to Product

The question: how do you make regime-dependent market behavior legible without flattening it into a snapshot? Static dashboards already do snapshots. We wanted users to watch correlation networks compress and fragment under stress, then drill into one asset at a single frozen moment. Two linked layers. Layer 1 is a scroll-driven force graph: 8 assets reacting to 25 years of rolling correlation. Layer 2 is an asset dossier that freezes time at the scroll position and reports price, volatility, drawdown, and that asset's cross-asset context. The welcome page sets the tone, the geographic globe grounds it.

What Changed Across Milestones

M1 fixed the dataset and the question. 8 instruments (4 commodities, 2 precious metals, 1 currency, 1 equity index) over 6,410 trading days. The Pearson matrix surfaced two clusters and three shock regimes (2008, 2020, 2022). M2 shipped the skeleton: force graph, scroll timeline, regime classifier (STABLE / ELEVATED STRESS / SYSTEMIC SHOCK), KPI panel. Items 1-3 of the core MVP. M3 closed items 4-7: node-click → dossier, frozen line chart, correlation breakdown, volatility & drawdown bars. Plus most of the Extras. The dossier carries six panels around the frozen chart. The welcome page replaced the old cold entry with a short intro animation. The globe shipped with country-click context. Layer 1 picked up volumetric nodes, animated link pulses, differentiated negative-correlation rendering, the regime glitch, and clickable crisis waypoints.

M1 | 20 MAR

Goal & Data

8 assets / 6,410 obs. Two-layer product pitched. Pearson matrix and shock-regime EDA delivered.

M2 | 16 APR

Skeleton

Force graph live. Scroll timeline + regime classifier shipped. 3 of 7 core items done.

M3 | 29 MAY

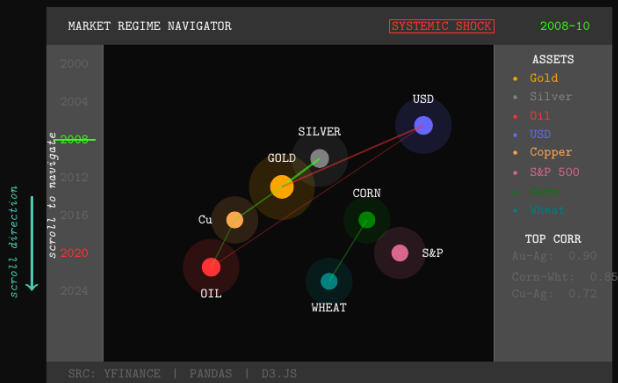
Final Product

Dossier, welcome page, globe shipped. Layer 1 rebuilt: neon nodes, link pulses, crisis waypoints, glitch.

MVP scoreboard. Core 7/7 shipped. M2: (1) force graph + scroll timeline, (2) regime classifier, (3) metrics panel. M3: (4) node-click → dossier, (5) frozen line chart, (6) correlation breakdown, (7) volatility & drawdown. Extras 4 of 5 shipped: producer/consumer globe; safe-haven gauge in dossier; regime-driven ambient audio on welcome page; regime transition effects as CSS glitch (particles dropped). Crisis annotations reshaped into clickable JUMP TO waypoints with tooltips.

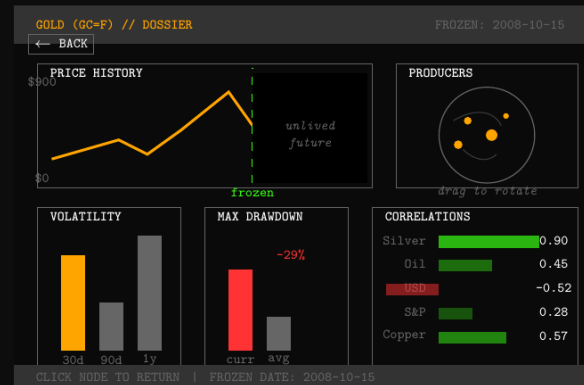
M1 / M2 Sketches

Layer 1: Main Dashboard



M1-era macro view. Force-graph wireframe with the scroll-as-time timeline on the left edge, a regime classifier in the header, and an ASSETS / TOP CORR panel on the right.

Layer 2: Asset Dossier



M2-era dossier wireframe. Frozen price chart with the unlived-future shading, PRODUCERS placeholder (became the globe page in M3), and volatility / drawdown / correlation panels along the bottom.

M1 fixed the two-layer split and the navigation arc: macro graph → frozen moment → asset dossier. M2 settled the layout conventions: scroll-as-time on the left edge, force graph as the canvas, and a three-state regime classifier naming the current regime. M3 kept both and added surface depth on top: volumetric nodes, animated link packets, the regime glitch, ambient audio, the multi-window volatility readout, the drawdown trajectory, and the geographic globe.

Motivation, Audience, Data Story

The audience is finance-curious students: readers who have followed news coverage of Lehman, COVID, and the 2022 rate shock but haven't seen the structure underneath - correlation matrices, the regime classifier, rolling volatility, drawdown, safe-haven behavior. The interface uses no β , no Sharpe, no Value-at-Risk. It shows how 8 assets are moving together at a given month, names the regime, and lets the user drop into one asset's experience of one moment.

Why static charts miss it. A heatmap shows one number at one moment, and you cannot index 25 years of them by hand. A price chart shows one asset over time and is blind to everything moving around it. Neither shows the property we care about: how the correlation network reshapes when stress hits. The cluster that holds in 2006 is not the cluster that holds in 2008, and Gold's correlation with the S&P 500 isn't a constant; it's a function of regime. In the force graph you watch this directly: links between assets tighten and loosen as you scroll, and clusters merge and split.

Data-story arc. Three beats. (1) See the system: scroll Layer 1 and watch correlation deform across 25 years. (2) Pick a moment with JUMP T0, or just scroll to park on any month. (3) Drill in. Clicking a node freezes time and opens the dossier, which carries the price line, volatility across three windows, drawdown, the per-asset correlation breakdown, the safe-haven gauge, and the globe that says where the asset is physically produced.

Layer 1 -- The Correlation Graph



M3 final. Live force graph at OCT 2008, regime SYSTEMIC SHOCK.

The M2 sketch fixed the layout: scroll-driven vertical timeline on the left, force-laid asset nodes in the center, color-coded links. M3 kept the layout and rewrote every panel inside it. Five upgrades shipped between M2 and M3. (1) Volumetric neon nodes replaced the flat SVG circles; each carries a radial-gradient glow whose intensity pulses with the asset's rolling volatility. (2) A JUMP TO crisis bar exposes 2008, 2020, 2022 as clickable waypoints, each with a hover tooltip that explains the crisis; clicking lights the matching year on the left timeline and autoscrolls into it. 2001 and 2002 also appear on the left timeline as flagged crisis ticks, but didn't get dedicated jump buttons because their structural signatures are subtler. (3) Behind the force layout sits an ambient lattice background. (4) The straight SVG links became curved quadratic paths, and small glowing data packets travel along positive correlations via animated dashed strokes. (5) Negative correlations are distinguished by color and glow, not by jagged geometry. The path itself is the same quadratic Bézier in both cases; the curvature offset just flips sign, and positive vs negative get separate stroke colors (green vs red) and separate glow filters.

A right-hand panel runs three things in parallel as the user scrolls: AVG COUPLING bars per asset, the current top correlation pairs, and a small TIME NAVIGATOR heatmap for direct month-jumps. On a STABLE → SYSTEMIC SHOCK crossover a glitch fires for ~600ms: a brief RGB-channel split on the header, a stutter on the regime badge, and a pulse on the edge glow. It triggers on cross only, not on revisit. Performance was the hard part. Each scroll position retargets the rolling 63-day correlation matrix (63 days ≈ one trading quarter, short enough that a regime change registers inside one window). The trick is that the correlation matrices are pre-computed once at build time, one per month-end, 304 in total across the 2000-09 to 2025-12 span, and shipped as a static JSON payload (correlation_data.json). On each scroll step the client looks up the nearest snapshot and gives the running D3 force simulation a small alpha kick (simulation.alpha(0.45).restart()); the sim relaxes into the new geometry over a few hundred ms. There is no browser-side correlation recomputation and no cached or interpolated node positions. Re-converging the sim each step turned out cheaper and smoother than interpolating between layouts.

Layer 2 -- The Asset Dossier



M3 final. Gold dossier, frozen at OCT 2008. DOTCOM and GFC shaded as inline crisis bands.

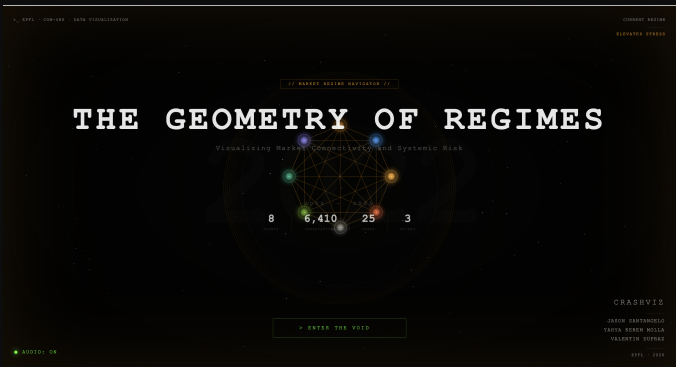
Clicking a node in Layer 1 freezes the timeline and opens the dossier. The M2 sketch carries through almost intact: frozen line chart top-left (63% wide) with the asset identity card top-right (37% wide, carrying AVG COUPLING, the safe-haven gauge, and the price/returns readouts); along the bottom, three panels in a 21 / 32 / 47 split: volatility, drawdown, correlation breakdown. The globe lives as a separate page reached from the Layer-1 node-click dossier panel, not inside the dossier grid. M3 added six elements.

- (1) The frozen line chart supports a regime-overlay mode that shades the background green/orange/red against the rolling classification, with inline crisis bands (DOTCOM, GFC) when the frozen date sits inside one.
- (2) A normalized base-100 toggle overlays any of the other 7 assets on the same scale.
- (3) AVG COUPLING AT FROZEN DATE gives the asset's mean correlation against the other 7 at the frozen moment (0.34 for Gold at OCT 2008).
- (4) The safe-haven gauge sits below the avg-coupling number and computes $\text{corr}(\text{asset}, \text{Gold}) - \text{corr}(\text{asset}, \text{S\&P 500})$: positive when the asset behaved like a hedge, negative when it tracked equities.
- (5) Volatility, three annualised windows: 30d, 63d, 252d.
- (6) The drawdown panel pairs the current-from-peak loss with the worst-ever-to-date loss, exposes both peak and trough dates, and adds a ZOOM TO CRASH action that zooms the line chart to the drawdown window.

The black void after the frozen date is the most contested design choice. At any historical moment the future was unknown, and rendering it as a dashed extrapolation would have implied otherwise, so the right edge stays a black void.

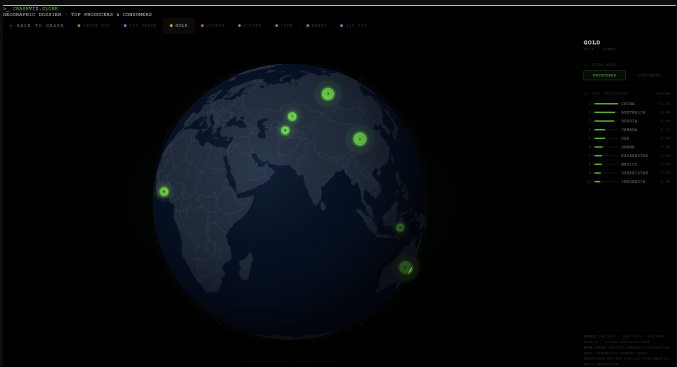
Every dossier metric is pre-computed at build time and shipped as static JSON. Pearson correlations, rolling volatility, max drawdown, and the safe-haven gauge are baked into `dossier_timeseries.json` and `price_series.json` rather than calculated client-side. Cold-load under 400ms.

The Welcome Page



Welcome page, post-entry. CURRENT REGIME reading top-right; asset constellation behind the title.

The Geographic Globe



CRASHVIZ.GLOBE. Top producers for Gold. Numbered dots rank by share of global output.

The welcome page was a late-stage addition that wasn’t sketched in M2. It replaces a cold dashboard entry with a short intro animation that sets the mood, skipping the usual onboarding pieces (animated illustrations, marketing copy, value-prop bullets). The page is a pitch-black void with a ghost-year watermark, CRT scanlines, a vignette, and an edge glow whose color is driven by the live regime classifier reading the most recent correlation matrix (ELEVATED STRESS at capture time). Behind the title sits an 8-node asset constellation, a static preview of the Layer 1 force graph. Four HUD corners hold the course tag, regime readout, audio toggle, and brand mark. The center runs a six-second entry: regime badge → title → subtitle → stat strip → > ENTER THE VOID.

Ambient audio. Opt-in; the toggle pulses green when active. No audio file. The sound is synthesized live via the Web Audio API. Four OscillatorNodes running sine waves layer into a chord, a fifth sine oscillator runs as an LFO tremolo, and a synthesized pink-noise channel adds rumble. The drone morphs with the live regime classifier: each regime carries its own profile (oscillator frequencies, LFO rate/depth, filter cutoff, noise level, volume), and crossover triggers a smooth `setTargetAtTime` morph over a few seconds. STABLE sits on a low C-major chord. ELEVATED STRESS pulls to unresolved tritone intervals with a faster tremolo and a darker filter. SYSTEMIC SHOCK drops to a diminished cluster (A1/D2/G2/C3) with heavy noise and a panicky tremolo. RECOVERY opens the filter and lifts into a brighter register. The audio is the same regime signal the screen carries, expressed as tone.

The M2 dossier sketch reserved space for a producer/consumer view as a small circle with three dots. M3 shipped that as a dedicated page (CRASHVIZ.GLOBE) reachable from the Layer-1 dossier panel: a D3-geo orthographic projection of the top producer and consumer nations for the selected asset. Asset selector along the top, PRODUCERS / CONSUMERS toggle on the right, ranked country list with shares, numbered green dots on the globe. Drag to rotate; click a country to lock its context.

Geographic data sources: USGS and the British Geological Survey for minerals (copper, silver, gold); the International Energy Agency and the Energy Institute Statistical Review for oil; BHP, Rio Tinto, and Glencore annual reports for production-attribution context; USDA for wheat and corn. The data layer was the longest single research effort in M3, and most of that time went into reconciling country naming across heterogeneous sources (USGS country labels, IEA regional groupings, USDA country codes, corporate-report jurisdictions) into a single ISO-aligned lookup that the globe’s COUNTRY_ISO table consumes at render time.

Design Decisions

Terminal aesthetic. Monospace throughout, dark background (#0D0D0D), teal accent (#4EC9B0). The reference is the Bloomberg terminal stripped to essentials.

Rejected: heatmap matrix. The standard quantitative-finance view of correlation, one cell per pair. Precise but static. A heatmap tells you two assets' correlation at one window, but not which cluster is holding, and 25 years of heatmaps stacked side-by-side is unreadable. The force graph trades precision for shape: any single chord is less precise than a heatmap cell, but the layout itself reads as a system, and the scroll timeline deforms it as you move through time. We did keep the heatmap where it works: the right-panel TIME NAVIGATOR is a small heatmap for direct month-jumps.

Rejected: data-journalism aesthetic. FT pink, NYT light grey, serif headlines, annotated lines. That palette reads calm and explanatory, which is the right tone for stable subject matter. Our subject is market panic. A calm interface would have undersold the data, so the dark/monospace pairing was the alternative we wanted: the screen and the regime read the same way.

Regime color. Green STABLE, orange ELEVATED STRESS, red SYSTEMIC SHOCK. The same triplet runs across the welcome badge, the force-graph header, the dossier line-chart overlay, timeline ticks, the welcome-page edge glow, and the audio profile, so the regime is one signal carried across every surface.

Tools & Lecture Mapping

D3 force simulation (link forces tuned by correlation magnitude, α -restart on scroll) → Graphs. D3-geo orthographic projection with drag-rotate, click-to-lock countries → Maps / Practical maps. SVG marks: quadratic-Bézier link paths, animated dashed strokes, radial-gradient nodes pulsing with volatility → D3.js + Marks & Channels. Color palette: three-state regime triplet (green/orange/red) reused across every surface → Perception of colors. Scroll-driven timeline + frozen-time dossier + JUMP TO chapters → Storytelling. Brush-and-link, focus-plus-context, hover tooltips, regime glitch → Interactions. Regime-reactive Web Audio drone → Sound viz / Music Viz. Python pre-computation pipeline (correlations, volatility, drawdown, safe-haven gauge baked into static JSON) → Data. Rejected-alternatives reasoning (heatmap, data-journalism aesthetic) → Design for data viz / Do's and don'ts.

What Was Hard

The first hurdle was data sync. NYSE, CBOT, and NYMEX run different calendars, which produced between 12 (USD Index) and 72 (Corn) NaN entries per asset across 25 years. Forward-fill carries the last valid close across non-trading days, residual head-of-timeline NaNs are dropped, and the April 20, 2020 negative WTI settlement is floored to 0.01 so rolling statistics stay stable.

Cross-page state contract. Three pages (index.html, dossier.html, globe.html, with the route back to welcome.html) share two pieces of state, the scroll-frozen month index and the selected asset, without a router or framework. The shipped contract is flat sessionStorage with two keys: mg_currentMonth and mg_lastAsset, plus the GLOBE_IDS table that bridges force-graph node IDs and dossier/globe asset IDs. We considered query-string routing but it would have added a parsing layer to every page.

The dossier grid is dense, and below ~1280px the panels don't reflow cleanly. We shipped desktop-first rather than dilute the experience; a mobile entry would need a separate swipeable card layout.

What We Cut

The M2 Extras list had five items. Three shipped (globe, safe-haven gauge, regime-driven audio), one reshaped (crisis annotations became clickable JUMP TO + tooltips), and one was dropped.

Particle effects on regime transitions got dropped in favor of the regime glitch (RGB split + text stutter). The particle prototype looked good but pulled focus from the node movement, which is the thing users should be watching.

We also cut an aggressive sting on every regime crossover. The shipped morph uses smooth setTargetAtTime ramps instead of a sting, because the sting prototype was jarring in practice.

Team Contributions

The three milestone reports (M1, M2, M3) are shared work, drafted and reviewed jointly. What follows is primary build ownership; every shipped surface received review and small fixes from at least one other team member.

Yahya Kerem Molla

Originated the project concept -- visualizing market structure under crises, the two-layer architecture, the 8-asset dataset -- and ran the project's general planning and coordination (milestone schedules, scope, sync). On the build, owned the geographic globe and the welcome page. Globe (globe.html). D3-geo orthographic projection; top-producer and top-consumer nodes per asset; drag-rotate; the click-to-lock country context (country name, role, share of global output in the right panel); the PRODUCERS / CONSUMERS toggle; the asset-selector tabs. The geographic data layer (USGS, BGS, IEA, Energy Institute Statistical Review, USDA, plus corporate annual reports for attribution context) was the longest single research effort of M3, mostly spent reconciling country naming across heterogeneous sources into the COUNTRY_ISO lookup. Welcome page (welcome.html). Landing layout, ghost-year typography, CRT scanlines and vignette, the regime-driven edge glow, the asset-constellation preview of Layer 1, the six-second entry sequence, and the regime-reactive Web Audio drone (four OscillatorNodes + sine-wave LFO + synthesized pink noise, morphing across four regime profiles via setTargetAtTime) with the opt-in audio toggle.

Jason Santangelo

Jason owned the asset dossier (Layer 2) and the M3 demo screencast. Dossier (dossier.html). compute_dossier_data.py pipeline producing the static JSON payloads (dossier_timeseries.json, price_series.json) for all 8 assets: pre-computed daily prices, Pearson correlations at every frozen date, three-window rolling volatility (30/63/252), max-drawdown trajectories with peak and trough dates, safe-haven gauge series. Grid layout and per-panel rendering: frozen line chart with regime-overlay shading and inline crisis bands, volatility bar trio, drawdown panel with ZOOM TO CRASH, correlation breakdown, safe-haven gauge. Chart interactions: tooltips, base-100 normalization overlay, the clickable interactions on the frozen line chart and risk-metric panels. Worked with Yahya on the dossier/globe handoff: both pages consume the shared two-key sessionStorage contract (mg_currentMonth, mg_lastAsset), so the user lands on the right asset and frozen date when navigating from the Layer-1 dossier panel into either page. The screencast scripts, records, and edits the M3 demo, covering the welcome-page entry, the Layer 1 scroll-through-crisis flow, the dossier drill-down, and the globe.

Valentin Dupraz

Owned index.html (Layer 1) and the GitHub repository. Layer 1 build. Volumetric neon nodes pulsing with rolling volatility; glowing crisis waypoints in the left timeline (clickable autoscroll into 2008, 2020, 2022); ambient lattice background; curved quadratic-Bézier tension links with animated dashed strokes and data-packet pulses for positive correlations; differentiated rendering of negative correlations (separate negative-link glow filter + opposite-curvature path); regime glitch (RGB channel split + text stutter) on STABLE → SYSTEMIC SHOCK crossover. Node-click dossier panel inside index.html. Slide-in side card opening on node click, populated with the asset's live correlations to all 7 others and an AVG COUPLING readout, with a regime-driven border color. A Layer-1 read-only mirror of the full dossier that lets the user inspect connectivity without leaving the force graph. sessionStorage handoff to Layer 2. The cross-page state contract: two keys (mg_currentMonth, mg_lastAsset) set on every leaveTo* navigation, read on load by both dossier.html and globe.html, plus the GLOBE_IDS table bridging force-graph node IDs and dossier/globe asset IDs. Help and tooltip system. The ? terminology popover, the per-panel info-i icons, the per-crisis tooltips on each JUMP TO waypoint, and the link-hover tooltips on the force graph. compute_correlations.py producing correlation_data.json (304 month-end 63-day rolling correlation matrices, 2000-09 to 2025-12) consumed by the Layer-1 force graph. On the repo side, Valentin set up and managed the GitHub: folder structure, README, file uploads, and the integration of the three team members' surfaces into a single deployable site.

AI tools used. Claude was used for code drafting, debugging, design exploration, and report proofreading. All AI-generated output was reviewed, tested, and verified by the team before merge or submission. Factual claims about the codebase were cross-checked against the source files.